MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

Model Question Paper for **MDC** (2024 ADMISION ONWARDS)

COURSE NAME- INDIAN MATHEMATICAL TRADITION, COURSE CODE – MG1MDCSAN100

Duration: 1.30 hrs. Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability. Answers may written either in Sanskrit / English or Malayalam.

Part A

Multiple Choice Questions Answer All Questions Each question carries 1 mark

1. This is one of the Vedanga वेदाङ्गेषु एक: भवति ।[Remember] [CO 1]

a) Jyotisha ज्योतिष b) Adharva अथर्व c) Ramayana रामायण d)Mahabharata महाभारत

2. Thousand in Sanskrit संस्कृत संख्या १००० ।[Remember] [CO 1]

a)अयुत b)सहस्र c)दश d)शत

3. This is one of the Yuga युगेषु एक: भवति।[Remember] [CO 2]

a) Kali कलि b)Kalpa कल्प c) Mahabharata महाभारत d) Adharva अथर्व

4. Sanskrit word for Four संस्कृत संख्या ४ ।[Remember] [CO 2]

a) सप्त b) चत्वारि c) एकम् d) अष्ट

5. 'What is the value of Ka क in 'Katapayadi कटपयादि system ? [Understant] [CO 3]

a) सप्त 7 b) चत्वारि 4 c) एकम् 1 d) अष्ट 8

6. Who is the author of Panchasiddhantika पञ्चसिधान्तिक ? [Understant] [CO 3]

a) Varahamihira वराहमिहिर b) Brahmagupta ब्रह्मगुप्त 4 c)Pingalacharya पिङ्गलाचार्य d) Bhaskaracharya भस्करचर्य

7. Name the work of Aaryabhata आर्यभट ? [Understant] [CO 2]

a) लीलावति b) बृहत्संहिता 4 c)पञ्चसिधान्तिक d) आर्यभटीय

8. बोधायानस्य काल: Age of Bhodhaayana? [Understant] [CO 2]

a) 19CAD b) 20AD 4 c)8BC-4BC d) 18CAD

9. How many vedas Vedanga वेदाङ्गा: कति ? [Understant] [CO 2]

a) 18 b) 8 c) 6 d) 16

10. How many chapters are in Aryabhateeya आर्यभटीय ? [Understant] [CO 2]

a) 18 b) 8 c) 6 d) 4

[1 Marks x 10 = 10]

Part B

Short Essay Type Questions

Answer 2 Questions

Each question carries 5 marks

11. Write a short essay on VedaGanita वेदगणितं उपन्यस्यत । [Analyse [CO1,3]

12.Write a short essay on Leelavati लीलावती अधिकृत्य उपन्यस्यत । [Analyse] [CO1]

13. Write a short essay on Arithmetics गणितम् अधिकृत्य उपन्यस्यत । [Analyse] [CO2]

14. Write a short essay on binary system of Pingalaacharya सङ्ख्याक्रमम् अधिकृत्य उपन्यस्यत । [Analyse] [CO3]

[5 Marks x 2 = 10]

Part C

Essay Type Questions

Answer 2 Questions Each question carries 15 marks

15. Write an essay on Sulba sutras & Indian Mathematical tradition

    शुल्बसूत्र भारतीय पारम्पर्यं अधिकृत्य उपन्यस्यत । [Analyse] [CO1]

16. 'Write an essay on 'Kerala mathematics tradition केरलीय पारम्पर्यं अधिकृत्य उपन्यस्यत । [Analyse] [CO2]

17. Write an essay on Bakshali manuscripts & Indian Mathematical tradition

    भाक्षाली हस्तलिखितादीन् भारतीय पारम्पर्यं अधिकृत्य उपन्यस्यत । [Analyse] [CO3]

18. Write an essay on Arithmetics & Algebra & Indian Mathematical tradition

    गणितं बीजगणितं भारतीय पारम्पर्यं अधिकृत्य उपन्यस्यत । [Analyse] [CO2]

[15 Marks x 2 = 30]